

SQL AGGREGATION FUNCTIONS ASSIGNMENT

ROUTE

ASSIGNMENT OVERVIEW

SQL Aggregation Functions, GROUP BY & HAVING:

This assignment focuses on SQL aggregation functions and how they are used with GROUP BY and HAVING to analyze and summarize data.

Instead of displaying individual rows, you will learn how to write queries that produce meaningful summaries, such as totals, averages, counts, and comparisons across different categories.

OBJECTIVES OF THIS ASSIGNMENT

By completing this assignment, you will be able to:

- Use aggregation functions such as
- COUNT(), SUM(), AVG(), MAX(), and MIN()
- Group data correctly using GROUP BY
- Filter aggregated results using HAVING
- Understand the difference between WHERE and HAVING
- Write correct and well-structured analytical SQL queries

HOW TO WORK ON THIS ASSIGNMENT

- Load the provided dataset into your database
- Read each question carefully
- Identify whether GROUP BY and/or HAVING is required
- Focus on understanding the logic behind each query

QUESTIONS

Open the provided link and download the dataset files.

https://drive.google.com/file/d/16KdsnhdpD_Q2PSDfO1f27VmqqvePPMTD5/view?usp=drive_link

Questions:

A- Basic Questions:

- 1- Query the total number of records.
 - 2- Query the average grade of all students.
 - 3- Query the highest grade obtained.
 - 4- Query the lowest grade obtained.
 - 5- Query the number of distinct students.
 - 6- Query the average grade for each course.
 - 7- Query the number of students in each department.
 - 8- Query the maximum grade for each course.
 - 9- Query the minimum grade for each department.
 - 10- Query the number of courses taken by each student.
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B- Intermediate Questions:

- 11- Query courses where the average grade is greater than 75.
 - 12- Query students who have taken more than 3 courses.
 - 13- Query departments where the average grade is less than 65.
 - 14- Query students whose maximum grade is 100.
 - 15- Query courses that have been taken by at least 20 students.
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C- Advanced Questions:

- 16- Query students who have taken at least 4 courses and whose average grade is greater than 80.
- 17- Query departments where the maximum grade is at least twice the average grade.
- 18- Query courses whose average grade is higher than the overall average grade.
- 19- Query students whose lowest grade is greater than 70.
- 20- Query students whose lowest grade is greater than 70.